

REPORT

Title: Fine Particulate Matter Exposure and Public Health Outcomes in Washington State

Authors: Sophia Patrin, Melodie Nekoorad

Summary of Research Questions and Results:

1. Is there a statistically significant correlation between county-level annual mean PM2.5 concentration and hospitalization rates for respiratory disease in Washington State in 2022?

Result: There is no statistically significant correlation between PM2.5 concentration and asthma prevalence. The correlation coefficient is close to zero, indicating no meaningful linear relationship between the two variables.

2. Do counties with the highest PM2.5 concentrations have significantly different asthma prevalence rates than other Washington counties?

Result: No statistically significant difference was found between the top 10 highest-PM2.5 counties and the remaining counties. The difference in asthma prevalence between the groups is small and not statistically significant.

3. Can county-level PM2.5 concentrations predict respiratory health outcomes using confidence intervals?

Result: PM2.5 concentration is not a statistically significant predictor of asthma prevalence. The regression model explains very little variation in asthma rates, and the confidence interval includes zero.

Motivation:

As climate change intensifies, wildfires have become more frequent and severe, particularly in Washington State. Wildfire smoke introduces high concentrations of fine particulate matter (PM2.5), along with harmful gases and toxins, posing serious public health risks. Exposure to PM2.5 is associated with respiratory illnesses, cardiovascular complications, and long-term health effects.

We chose to focus on the 2022 wildfire season due to its severity and prolonged impact in

Washington State. As residents of the Pacific Northwest, this issue is highly relevant to our communities and directly affects public health outcomes in the region. Understanding the relationship between air pollution and respiratory health is essential for informing public health strategies and environmental policy.

Datasets:

Dataset 1: EPA Air Quality System (AQS) Daily PM2.5 Data, 2022 - [Download Daily Data | US EPA](#)

This dataset contains daily PM2.5 measurements for Washington State in 2022. We aggregated daily values into annual averages at the county level using the *Daily Mean PM2.5 Concentration* column.

Dataset 2: CDC PLACES: Local Data for Better Health, County Data 2024 Release - [PLACES: Local Data for Better Health, County Data 2024 release | Data | Centers for Disease Control and Prevention](#)

This dataset provides model-based estimates of health indicators based on 2021–2022 survey data. We filtered the data for Washington State and used the *CASTHMA* variable (adult asthma prevalence) as our outcome variable.

Note: CDC PLACES provides model-based estimates rather than direct measurements, so we pair it with 2022 EPA data to maintain temporal consistency with the 2021–2022 survey period. For this analysis, we used crude prevalence estimates from CDC data.

Method:

Research Question 1:

To investigate whether PM2.5 concentration and asthma prevalence were correlated, we first cleaned and aggregated the EPA PM2.5 dataset to county-level annual means. We then cleaned the CDC asthma dataset and merged the datasets by county name. Using the SciPy library, we computed the Pearson correlation coefficient and corresponding p-value between annual mean PM2.5 concentration and adult asthma prevalence.

Research Question 2:

To compare counties with the highest PM2.5 concentrations against the remaining counties, we sorted counties by annual mean PM2.5 concentration in descending order and selected the top 10 counties. We then compared asthma prevalence rates between the top 10 highest-PM2.5 counties, and all remaining counties. Then, an independent two-sample t-test was performed using SciPy to determine whether the difference in asthma prevalence between the groups was statistically significant. A bar chart comparing mean asthma prevalence between groups was also generated, and includes standard deviation error bars.

Research Question 3:

To determine whether PM2.5 concentrations could predict asthma prevalence, we used linear regression with annual mean PM2.5 concentration as the independent variable, and adult asthma prevalence as the dependent variable. We used the statsmodels library to fit an OLS regression model. The model reports an r-squared value, p-value, and 95% confidence intervals. A regression visualization was produced using seaborn's regplot function.

EDA Summary:

During the exploratory data analysis, we identified important characteristics and limitations of both of our datasets. We examined dataset size, missing values, data types, and summary statistics. We identified and removed a few non-meaningful values from the PM 2.5 column, since negative pollution concentrations are the result of instrument malfunction. We also verified that the CDC asthma prevalence dataset contained no missing values in the variables relevant to our analysis.

Visualizations and summary statistics revealed important characteristics of the data. PM2.5 concentrations were strongly right-skewed, with occasional extreme spikes likely associated with wildfire events during late summer and early fall, as seen in **Figure 1**. In contrast, asthma prevalence was relatively consistent across Washington counties, with a roughly symmetric distribution and few outliers.

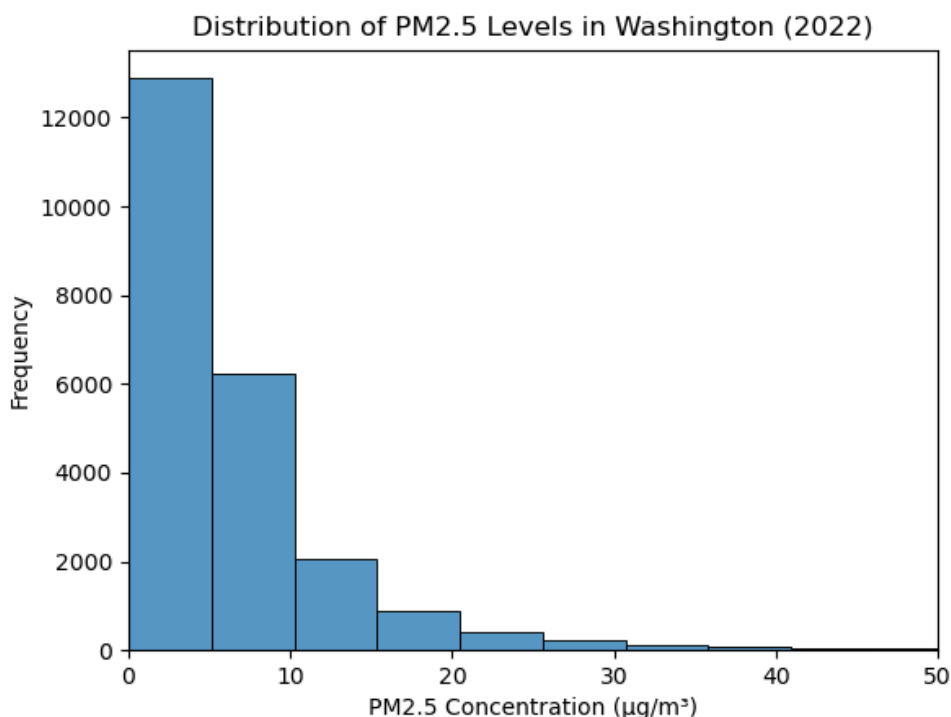


Figure 1: Distribution of daily PM2.5 concentrations showing most values are low, with a right-skewed tail representing extreme pollution events.

This histogram displays the distribution of daily PM2.5 concentrations across Washington State in 2022. A histogram was chosen to visualize how frequently different pollution levels occur. The plot shows that most PM2.5 values are concentrated at lower levels (below 10 $\mu\text{g}/\text{m}^3$), while a small number of extreme values extend far to the right. These high values indicate occasional significant pollution events, which are likely associated with wildfire smoke.

For the CDC asthma dataset, exploratory analysis showed that asthma prevalence was relatively consistent across Washington counties. Unlike the PM2.5 data, asthma prevalence appeared approximately symmetric with low variability and few outliers.

From the EDA we learned that eight Washington counties (Columbia, Ferry, Garfield, Pend Oreille, San Juan, Skamania, Stevens, and Wahkiakum) didn't have any EPA PM 2.5 monitoring data in 2022 and therefore could not be included in the final analysis which is a significant limitation. After

merging datasets, the final dataset contained 31 counties.

Finally, scatterplots generated during EDA suggested little visible linear relationship between county-level PM2.5 concentrations and asthma prevalence. This observation helped guide our later statistical analyses and foreshadowed the results of the correlation and regression models.

Results:

Research Question 1:

The Pearson correlation analysis found a very weak relationship between annual mean PM2.5 concentration and adult asthma prevalence ($r = -0.07$, $p = 0.60$). Because the p-value exceeds the significance threshold of 0.05, we fail to reject the null hypothesis that no relationship exists between these variables.

The correlation coefficient is extremely close to zero, indicating that counties with higher PM2.5 concentrations did not consistently experience higher asthma prevalence rates. The scatterplot further supports this conclusion, as the line of best fit is nearly horizontal and there is no visible trend in the data. These findings suggest that annual county-level PM2.5 exposure alone does not explain variation in asthma prevalence across Washington counties.

Research Question 2:

We compared asthma prevalence in the ten counties with the highest annual PM2.5 concentrations to the remaining counties using an independent two-sample t-test. The results indicated that the difference in asthma prevalence between the two groups was not statistically significant ($t = -0.734$, $p = 0.4790$). Although several high-pollution counties showed slightly elevated asthma prevalence, the overall difference between groups was small.

As shown in **Figure 2**, the mean asthma prevalence for both groups is nearly identical, and the error bars show substantial overlap, indicating high variability within groups. Although some high-PM2.5 counties exhibit slightly higher asthma rates, the overall difference is minimal.

These results suggest that counties with higher PM2.5 exposure do not consistently experience worse asthma outcomes. This indicates that grouping counties solely by average pollution levels may not capture the primary drivers of asthma prevalence.

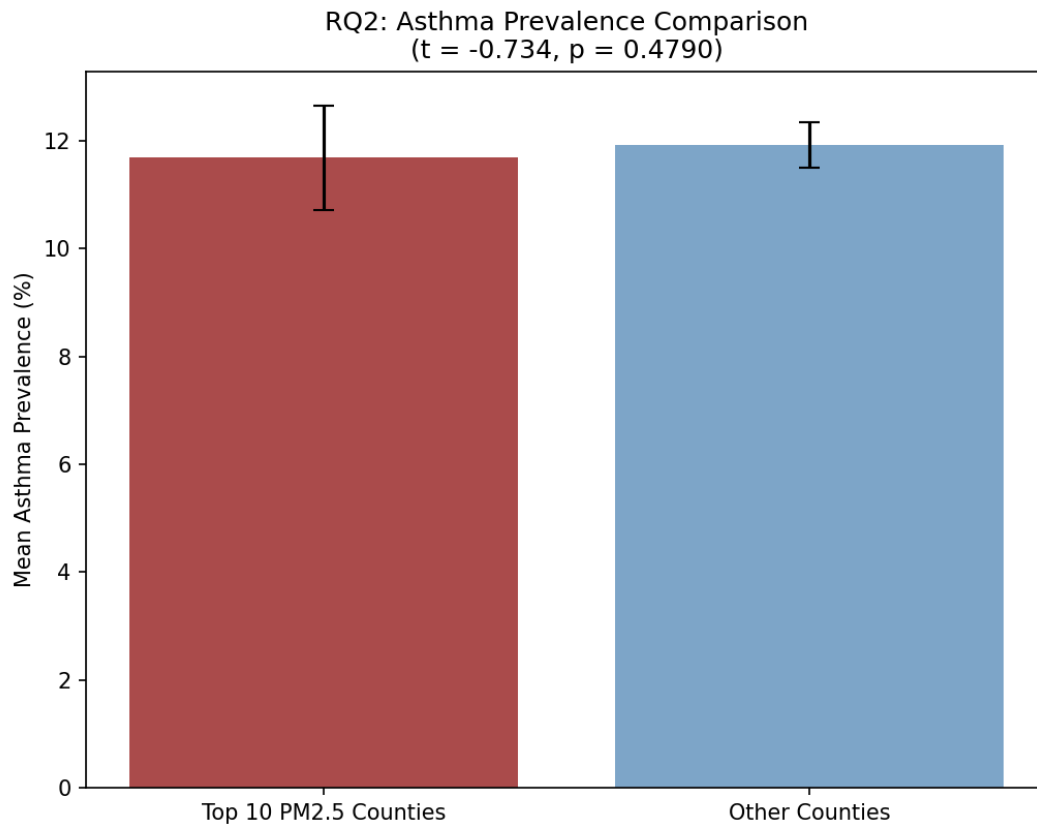


Figure 2: Comparison of mean asthma prevalence between the top 10 highest-PM2.5 counties and all other counties. Error bars represent variability within each group. The difference is not statistically significant.

Research Question 3:

The linear regression analysis demonstrated that PM2.5 concentration is not a statistically significant predictor of asthma prevalence ($p = 0.68$). The regression coefficient is small, and the 95% confidence interval includes zero, indicating no reliable effect. Additionally, the model produced a very low R^2 value, meaning that PM2.5 explains very little variation in asthma prevalence.

As shown in **Figure 3**, the regression line is nearly flat, and the data points are widely dispersed. The wide confidence interval reflects substantial uncertainty in the relationship. These findings indicate that PM2.5 concentration is not a useful standalone predictor of asthma prevalence in this dataset.

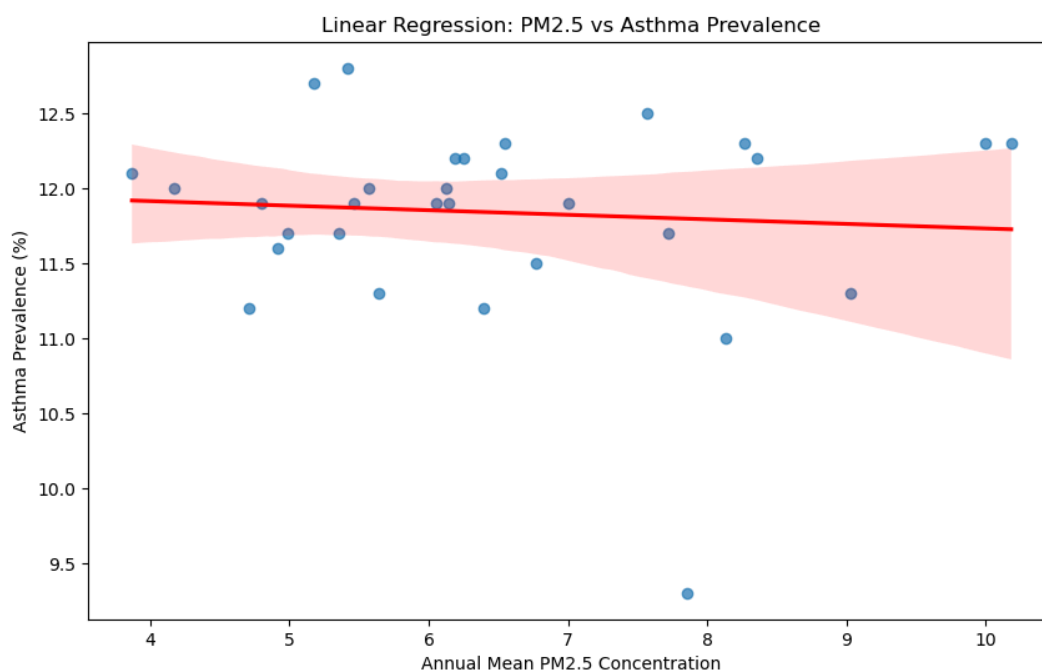


Figure 3: Linear regression of PM2.5 concentration versus asthma prevalence. The model shows a weak relationship with high variability and no statistical significance.

Across all three research questions, the results consistently indicate that PM2.5 concentration alone does not strongly influence county-level asthma prevalence. This suggests that asthma outcomes are likely influenced by additional factors beyond average pollution exposure.

Impact and Limitations:

Although this analysis did not find statistically significant relationships between PM2.5 concentration and asthma prevalence, the results highlight the complexity of environmental health outcomes and the challenges of using aggregated data to study them.

One important limitation arises from the use of county-level averages. Aggregating PM2.5 exposure at the county level may mask substantial variation within counties and fails to capture individual-level exposure differences. Residents within the same county may experience very different levels of pollution depending on proximity to pollution sources such as highways or wildfire smoke.

plumes.

Additionally, the use of annual average PM_{2.5} concentrations may obscure the health effects of short-term exposure spikes. Wildfire smoke events often produce extremely high PM_{2.5} levels over short periods of time, which may have acute health impacts that are not visible when averaged over an entire year.

Another limitation is the use of CDC PLACES data, which provides model-based estimates of asthma prevalence rather than direct clinical or hospitalization records. While these estimates are useful for large-scale analysis, they introduce uncertainty and may not fully capture true disease prevalence.

Data availability also constrained the analysis. Eight Washington counties lacked PM_{2.5} monitoring data and were excluded, reducing the sample size and limiting geographic representation. These counties tend to be more rural, potentially introducing bias into the dataset.

Furthermore, asthma prevalence reflects long-term health patterns influenced by a combination of environmental, genetic, and socioeconomic factors. As a result, asthma outcomes may not respond immediately to short-term changes in air quality within a single year of data.

Importantly, these findings should not be interpreted as evidence that PM_{2.5} exposure has no impact on respiratory health. Extensive prior research demonstrates that fine particulate matter is harmful. Instead, our results indicate that annual county-level averages alone are insufficient to capture this relationship, and that more detailed data is required.

Future research could address these limitations by incorporating more detailed and comprehensive data. Using individual-level or census tract-level data would provide a more accurate representation of exposure and health outcomes. Finer geographic resolution would help identify localized patterns that are lost in county-level aggregation. Additionally, future studies could include additional explanatory variables, such as socioeconomic status, access to healthcare, smoking prevalence, or population density. These factors are known to influence respiratory health and may help explain the variation in asthma prevalence observed across counties. Another potential extension would involve incorporating hospitalization or emergency room visit data for respiratory conditions, which may be more sensitive to short-term changes in air quality than long-term prevalence measures.

Challenge Goals

Result Validity:

For each research question, statistical significance testing was used to evaluate whether observed relationships are meaningful. We used SciPy to compute the Pearson correlation coefficients, p-values, and Welch's t-tests.

Assumption checking: Before performing parametric tests (Pearson correlation, t-test, linear regression), we verified key assumptions:

Normality: Asthma prevalence data showed roughly normal distribution

Independence: Counties are geographically distinct

Homoscedasticity: Residual plots showed no clear pattern

Statistical significance: We used $\alpha = 0.05$ as our significance threshold, consistent with standard practice.

Confidence Intervals: For regression analysis, we reported 95% confidence intervals, which include zero for the PM2.5 coefficient, confirming a lack of predictive power.

Multiple Testing Consideration: We performed only three hypothesis tests (correlation, t-test, regression), minimizing comparison bias.

New Library:

This project used both SciPy and statsmodels, which were not covered in CSE 163 coursework. SciPy was used for statistical hypothesis testing and analyzing correlation, while statsmodels was used to perform OLS linear regression, generate confidence intervals, and regression summaries.

Work Plan Evaluation:

Our original work plan estimates were relatively accurate, although certain stages required more time than initially expected. The data cleaning and merging sections from the exploratory data analysis took a lot longer than predicted. The issue here was we worked independently on each dataset, and they struggled to merge and debug the code asynchronously and we were less time efficient when doing our independent parts separately, rather than meeting and working together. Collaboration improved once we began working together and using GitHub more effectively to coordinate changes and integrate code.

Testing:

In order to ensure our results, we created a separate testing file containing tests for our data cleaning and statistical analysis functions using assert statements. For the merged dataset, we tested that the dataframe successfully loaded and included the required columns: county, annual_mean_pm25, and asthma_prevalence. We also verified that there were no missing values in the PM2.5 or asthma prevalence columns after cleaning and merging the datasets.

To validate the EPA air quality data, we tested that all PM2.5 concentration values were nonnegative, since negative pollution values would be impossible and would indicate a data cleaning error. Similarly, we tested that asthma prevalence values from the CDC dataset were nonnegative.

We additionally tested that our merged Washington State dataset contained county-level observations after merging, ensuring that the join between the EPA and CDC datasets was successful.

Finally, we tested that each research question function (correlation analysis, t-test analysis, and linear regression analysis) executed successfully without producing runtime errors. This helped confirm that the statistical methods and visualizations were functioning correctly on the cleaned dataset.

Together, these tests increased our confidence that the datasets were processed correctly and that the statistical results presented in our report were generated consistently and accurately.

Citations and Collaboration:

This project was completed collaboratively by the authors. External resources were used for background research and statistical reference.

References

- Jaffe, D. A., O'Neill, S. M., Larkin, N. K., Holder, A. L., Peterson, D. L., Halofsky, J. E., & Rappold, A. G. (2020). Wildfire and prescribed burning impacts on air quality in the United States. *Journal of the Air & Waste Management Association*, 70(6), 583–615.
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